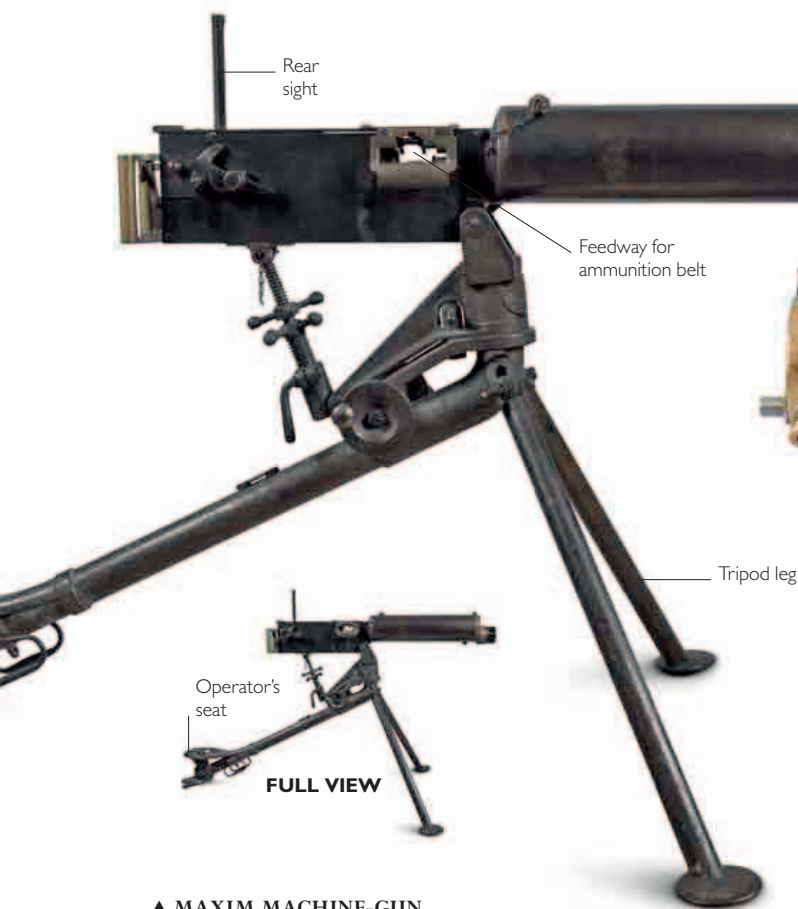


RECOIL-OPERATED MACHINE-GUNS (1896–1917)

Machine-guns built at the turn of the 20th century were either recoil-operated or gas-driven (see pp. 192–93). They were produced in Europe when the continent was devoid of conflict. Materials such as brass were plentiful and were used to make gun parts such as water jackets and spring housings. As Europe entered World War I, brass became scarcer, and steel—which was less expensive and also more durable—began to be employed for making gun parts. Gas-operated machine-guns could withstand greater pressure than recoil-operated ones and fire more powerful ammunition. However, recoil-operated machine-guns were more common, because their simple, reliable design found greater favor with troops.



▲ MAXIM MACHINE-GUN MODEL 1904

Date 1904

Origin UK

Barrel 28½in (72.3cm)

Caliber .30in-03

The Maxim gun was robust in construction and almost foolproof in design, taking on many forms after its introduction, including this upgraded model. This Maxim gun was the first rifle-caliber machine-gun formally adopted into US service. It was manufactured in .30in-03 caliber in the US by British gun manufacturer Vickers, Sons and Maxim. Later, some units were manufactured under licence in the US by Colt. Eventually, most Maxim models began to accept the newly introduced US .30in-06 cartridge. This gun fired 400–600 rounds per minute.



► BROWNING MODEL 1917

Date 1917

Origin US

Barrel 24in (61cm)

Caliber .30in

John Browning produced a poorly designed machine-gun for Colt, the gas-operated M1895 “Potato Digger” (see p. 194), but later reverted to recoil operation when creating the M1917. It fired 400–520 rounds per minute. This gun sported a water jacket to cool its barrel during operation. Water-cooled machine-guns usually required a two-man crew—one to fire the gun and the other to monitor the jacket and the fabric belt which loaded ammunition into the gun. Ammunition belts could carry a higher number of cartridges than standard magazines and they could be loaded easily via the feedways on these guns.

Foot of tripod leg fitted with cleat to stabilize the gun by preventing movement

